

# Operations on functions

Building new functions out of old ones, reading a graph, and the rules of logarithms.

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# Adding and multiplying functions

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 **Practise:** Quotients and sums of fractions · Add & subtract rational expressions ([https://www.khanacademy.org/math/prec calculus/x9e81a4f98389efdf:rational-functions/x9e81a4f98389efdf:adding-and-subtracting-rational-expressions/e/adding\\_and\\_subtracting\\_rational\\_expressions\\_5](https://www.khanacademy.org/math/prec calculus/x9e81a4f98389efdf:rational-functions/x9e81a4f98389efdf:adding-and-subtracting-rational-expressions/e/adding_and_subtracting_rational_expressions_5))

 **Practise:** Products of functions · Multiply binomials (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratics-multiplying-factoring/x2f8bb11595b61c86:multiply-binomial/e/multiply-binomials-coefficient>)

Khan has no lesson on  $f + g$ ,  $f - g$  and  $fg$  as such. These two are the same algebra on expressions, which is the closest it gets.

## THE FIVE

$$(f + g)(x) = f(x) + g(x) \quad (fg)(x) = f(x)g(x)$$

and the same idea for the difference, the quotient, and a constant times a function.

- 1 **Do the operation one value at a time.** Whatever  $x$  is, work out  $f(x)$  and  $g(x)$ , then combine those two numbers.
- 2 **The result is defined only where both are.** Intersect the two domains.
- 3 **A quotient also needs the bottom to be non-zero.**  $(f/g)(x)$  needs  $g(x) \neq 0$ .
- 4 **The combination you will use constantly is profit.**

$$P(q) = R(q) - C(q)$$

- 5 **Cost usually splits into fixed plus variable.** The fixed part is what you pay at  $q = 0$ , so it is the  $y$ -intercept.

## The Rivella Coffee case

### The five operations

$$(f + g)(x) = f(x) + g(x) \quad (f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x) \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad (cf)(x) = c \cdot f(x)$$

WORKED, WITH  $F(X) = 3X - 1$  AND  $G(X) = X^2 + 3X$

$$(f + g)(x) = x^2 + 6x - 1$$

$3x - 1 + x^2 + 3x$ , collected.

$$(f - g)(x) = -x^2 - 1$$

$3x - 1 - x^2 - 3x$ . The  $3x$  terms cancel.

$$(fg)(x) = 3x^3 + 8x^2 - 3x$$

$(3x - 1)(x^2 + 3x)$ , expanded.

$$\left(\frac{f}{g}\right)(x) = \frac{3x - 1}{x^2 + 3x}$$

Undefined where  $x^2 + 3x = 0$ , so at 0 and  $-3$ .

$$\left(\frac{1}{2}f\right)(x) = \frac{3}{2}x - \frac{1}{2}$$

A constant multiplies every term.

### Why it matters: profit

$$P(q) = R(q) - C(q)$$

#### WATCH OUT

- Put the second function in brackets before subtracting, or the signs of its later terms survive when they should not.
- A quotient is undefined wherever the bottom function is zero, even where the top is perfectly fine.
- The combination is only defined where both originals are. Intersect the domains.

# Reading a graph

KHAN ACADEMY

► **Watch:** Both intercepts · x-intercepts and y-intercepts (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/v/introduction-to-intercepts>)

► **Watch:** Intercepts of a parabola · Interpret a quadratic graph (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:quadratic-functions-equations/x2f8bb11595b61c86:intro-parabolas/v/interpret-a-quadratic-graph>)

≡ **Read:** Intercepts of lines review (x-intercepts and y-intercepts) (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/a/intercepts-of-lines-review-x-intercepts-and-y-intercepts>)

✍ **Practise:** From the formula · Intercepts from an equation ([https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/e/solving\\_for\\_the\\_x-intercept](https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/e/solving_for_the_x-intercept))

✍ **Practise:** Intercepts from a graph (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:linear-equations-graphs/x2f8bb11595b61c86:x-intercepts-and-y-intercepts/e/linear-function-intercepts>)

- 1 **The y-intercept is the value at zero.** Put  $x = 0$ . There is at most one.
- 2 **The x-intercepts are where the function is zero.** Solve  $f(x) = 0$ . There may be several, or none.
- 3 **Plot the intercepts first, then enough points to see the shape.**
- 4 **In a business question the intercepts mean something.** The y-intercept is usually a fixed cost. An x-intercept is usually break-even.
- 5 **A vertical line is not a function.**  $x = 3$  is a perfectly good line and has no y to be a function of x.

## The method

**WORKED:**  $Y = 2X + 3$

$$y = 0 \rightarrow 0 = 2x + 3$$

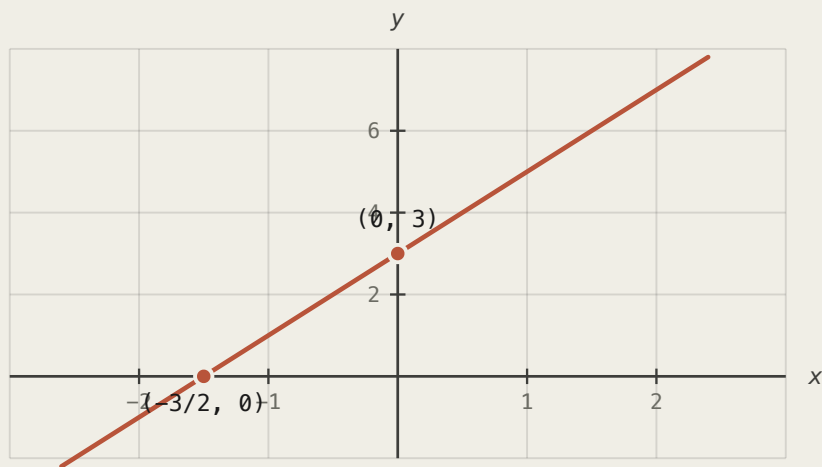
Set  $y$  to zero for the  $x$ -intercept.

$$x = -\frac{3}{2}$$

One  $x$ -intercept.

$$x = 0 \rightarrow y = 2(0) + 3 = 3$$

Set  $x$  to zero for the  $y$ -intercept.



$y = 2x + 3$ . It crosses the vertical axis at 3 and the horizontal axis at  $-3/2$ .

**A SECOND:**  $x = 3$

$$x = 3$$

A vertical line.

**no  $y$ -intercept**

Because  $x$  can never be 0 on this graph.

## Break-even is an $x$ -intercept

### WATCH OUT

- A zero denominator is not an  $x$ -intercept. It is a break in the curve.
- For a quadratic, a negative discriminant means no  $x$ -intercept at all, and that is a complete answer.
- Only the numerator can make a fraction zero, so solve the numerator and then check the value is allowed.

# Composition of functions

KHAN ACADEMY

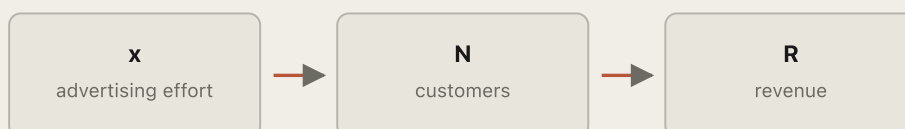
- ▶ **Watch:** What a composite is · Composing functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/v/function-composition>)
- ▶ **Watch:** At a number · Evaluating composite functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/v/evaluating-composite-functions>)
- ≡ **Read:** The definition · Intro to composing functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/a/introduction-to-function-composition>)
- ✍ **Practise:** At a number · Evaluate composite functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/e/evaluate-composite-functions-from-formulas>)
- ✍ **Practise:** Building the formula · Find composite functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:composing/e/compose-functions>)

## COMPOSITION

$$(g \circ f)(x) = g(f(x))$$

Do  $f$  first, then feed its output straight into  $g$ . The circle is read right to left.

- 1 **Work from the inside out.** Compute the inner value, then put it into the outer function.
- 2 **Substitute the whole inner output.** Every  $x$  in the outer function becomes the entire inner expression, brackets included.
- 3 **Order matters.**  $f \circ g$  is almost never  $g \circ f$ . Check with one number if you doubt it.
- 4 **For it to exist, the middle value has to be allowed.**  $f(x)$  must be something  $g$  accepts.
- 5 **The quantity in the middle is real, but the question is not about it.** Advertising gives customers, customers give revenue; the question asks revenue from advertising.



The chain. The middle quantity is real, but the question is not about it.

#### THE CALCULATION

$$R(N(x)) = 70(100\sqrt{x}) + \frac{(100\sqrt{x})^2}{20}$$

Every  $n$  becomes  $100\sqrt{x}$ , including the one being squared.

$$= 7000\sqrt{x} + 500x$$

Because  $(100\sqrt{x})^2 = 10\,000x$  and  $\frac{10\,000x}{20} = 500x$ .

#### The definition

$$(g \circ f)(x) = g(f(x))$$

#### Two properties

ORDER MATTERS:  $F(x) = 2x + 1$ ,  $G(x) = x^2 - 3$

$$(g \circ f)(x) = (2x + 1)^2 - 3 = 4x^2 + 4x - 2$$

Write the outer function with an empty bracket, then fill it.

$$(f \circ g)(x) = 2(x^2 - 3) + 1 = 2x^2 - 5$$

The other order.

**at  $x = 4$ : 78 against 27**

Same functions, same input, different answers.

#### WATCH OUT

- $(g \circ f)(x)$  is not  $g(x) \cdot f(x)$ . The ring is not a multiplication dot.
- The whole inner function goes inside the bracket:  $(2p + 1)^2$ , never  $2p^2 + 1$ .
- Take the domain from the two original functions, not from the tidied-up composite.

# Inverses, and when one exists

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► **Watch:** What an inverse is · Intro to inverse functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/v/introduction-to-function-inverses>)

► **Watch:** When one exists · Invertible functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:invertible/v/determining-if-a-function-is-invertible>)

≡ **Read:** One-to-one, written out · Intro to invertible functions (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:invertible/a/intro-to-invertible-functions>)

✍ **Practise:** Spotting when there is no inverse · Determine if a function is invertible (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:invertible/e/inverse-domain-range>)

✍ **Practise:** Reading an inverse off a graph · Inverse functions: graphs and tables (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:inverse-functions-in-graphs-and-tables/e/inverse-functions-graphs-and-tables>)

## INVERSE

The function that sends each output back to the input it came from.

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

## ONE-TO-ONE

Different inputs always give different outputs. No output is ever produced twice.

- 1 **A function has an inverse exactly when it is one-to-one.** That is the whole condition.
- 2 **On a graph, the horizontal line test decides.** A horizontal line may meet the graph at most once. Twice means two inputs share an output, so there is no way back.
- 3 **Strictly increasing or strictly decreasing is enough.** A function that only ever rises, or only ever falls, is one-to-one.
- 4 **The domain and the range swap.** The inverse's inputs are the original's outputs.
- 5 **The graph of the inverse is the reflection in the line  $y = x$ .**

WATCH OUT

- An inverse is not a reciprocal.  $f^{-1}(x)$  is not  $\frac{1}{f(x)}$ .
- A function that repeats an output has no inverse, because there is no way back to which input it came from.
- Swapping the domain and the range is part of the answer, not an afterthought.

$$Q(p) = \frac{50\,000\,000}{p+1} - 5\,000\,000 \quad \text{for } 0 \leq p \leq 9$$

#### THE DERIVATION

$$q = \frac{50\,000\,000}{p+1} - 5\,000\,000$$

Start from the definition.

$$q + 5\,000\,000 = \frac{50\,000\,000}{p+1}$$

Move the constant across.

$$p+1 = \frac{50\,000\,000}{q+5\,000\,000}$$

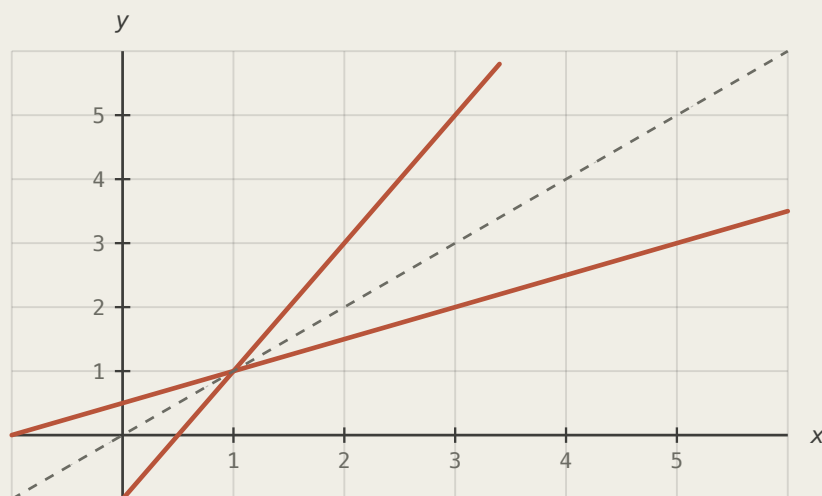
Take reciprocals of both sides.

$$P(q) = \frac{50\,000\,000}{q+5\,000\,000} - 1$$

The price as a function of the quantity.

#### The definition and the check

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$



$f(x) = 2x - 1$  and its inverse, mirrored in the dashed line  $y = x$ .

# Finding an inverse

KHAN ACADEMY

- ▶ **Watch:** Finding one, step by step · Finding inverse functions: linear (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/v/function-inverse-example-1>)
- ▶ **Watch:** Checking it · Verifying inverse functions by composition (<https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:verifying-inverse/v/verify-inverse-functions-from-tables>)
- ≡ **Read:** The method, written out · Intro to inverse functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/a/intro-to-inverse-functions>)
- ✍ **Practise:** Finding one · Find inverses of linear functions (<https://www.khanacademy.org/math/algebra/x2f8bb11595b61c86:functions/x2f8bb11595b61c86:inverse-functions-intro/e/algebraically-finding-inverses>)
- ✍ **Practise:** Checking by composition · Verify inverse functions ([https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:verifying-inverse/e/inverses\\_of\\_functions](https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:composite/x9e81a4f98389efdf:verifying-inverse/e/inverses_of_functions))

- 1 Write  $y = f(x)$ .
- 2 Solve for  $x$  in terms of  $y$ . This is the only step that is real work.
- 3 Swap the names.  $x$  goes back to being the input, so the answer is written as a function of  $x$ .
- 4 Check by composing both ways. You should get  $x$  back each time. This catches almost every slip.
- 5 If it is not one-to-one, restrict the domain until it is.  $f(x) = (x - 1)^2$  has no inverse until you keep only  $x \geq 1$ .
- 6 The linear case is worth knowing by sight.

$$f(x) = ax + b \quad \Rightarrow \quad f^{-1}(x) = \frac{x - b}{a}$$

- 7 To solve  $f(x) = k$ , apply the inverse to both sides. That is what an inverse is for.

## Finding one, and when it exists

THE GENERAL LINEAR CASE:  $F(x) = Ax + B$

$$g(x) = \frac{x - b}{a}$$

Undo in reverse order: subtract b, then divide by a.

$$f(g(x)) = a \frac{x - b}{a} + b = x$$

Check one way.

$$g(f(x)) = \frac{(ax + b) - b}{a} = x$$

And the other. So it is the inverse, and it is linear too.

## Using an inverse to solve an equation

### WATCH OUT

- $f^{-1}(x)$  is the inverse function, not  $\frac{1}{f(x)}$ .
- A function that fails the horizontal line test has no inverse until its domain is restricted.
- Domain and range swap. If you are asked for the domain of the inverse, look at the range of the original.

# The rules of logarithms

KHAN ACADEMY

- ▶ **Watch:** The logarithm rules · Intro to logarithm properties (1 of 2) (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/v/introduction-to-logarithm-properties>)
- ▶ **Watch:** The rule that solves equations · Using the logarithmic power rule (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/v/logarithm-of-a-power>)
- ▶ **Watch:** The change of base formula for logarithms (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:change-of-base/v/change-of-base-formula>)
- ≡ **Read:** The rules, written out · Intro to logarithm properties (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/a/properties-of-logarithms>)
- ✍ **Practise:** Expanding and condensing · Use the properties of logarithms ([https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/e/logarithms\\_2](https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:log-prop/e/logarithms_2))
- ✍ **Practise:** Evaluate logarithms: change of base rule (<https://www.khanacademy.org/math/algebra2/x2ec2f6f830c9fb89:logs/x2ec2f6f830c9fb89:change-of-base/e/evaluate-logarithms-using-the-change-of-base-rule>)

## THE THREE THAT EXIST

$$\log(ab) = \log a + \log b \quad \log \frac{a}{b} = \log a - \log b \quad \log(a^n) = n \log a$$

## THE ONE THAT DOES NOT

There is no rule for  $\log(a + b)$ . It does not simplify. Nothing you do turns a sum inside a logarithm into anything shorter.

- 1 **Expanding:** a product becomes a sum, a quotient a difference, a power a coefficient.
- 2 **Simplifying:** the same three rules read right to left.
- 3 **The power rule is the one that solves equations.** It brings the unknown down out of the exponent where you can reach it.
- 4 **Change the base when the calculator has not got it.**

$$\log_b x = \frac{\ln x}{\ln b}$$

5 The exponential and the logarithm undo each other.

$$\exp(\ln u) = u \quad a^x = \exp(x \ln a) \quad \text{for } a > 0$$

6 Check the argument is positive before applying any of it.

### Exponents

| RULE                                | RULE                        |
|-------------------------------------|-----------------------------|
| $b^x \cdot b^y = b^{x+y}$           | $\frac{b^x}{b^y} = b^{x-y}$ |
| $(b^x)^y = b^{xy}$                  | $(bc)^x = b^x c^x$          |
| $(\frac{b}{c})^x = \frac{b^x}{c^x}$ | $b^1 = b$                   |
| $b^0 = 1$                           | $b^{-x} = \frac{1}{b^x}$    |

### Logarithms

| RULE                                   | RULE                                       |
|----------------------------------------|--------------------------------------------|
| $\log_b(mn) = \log_b m + \log_b n$     | $\log_b \frac{m}{n} = \log_b m - \log_b n$ |
| $\log_b(m^r) = r \log_b m$             | $\log_b \frac{1}{m} = -\log_b m$           |
| $\log_b 1 = 0$                         | $\log_b b = 1$                             |
| $\log_b m = \frac{\log_a m}{\log_a b}$ | change of base                             |

## Expanding

### WRITING LOGARITHMS IN TERMS OF SIMPLER LOGARITHMS

$$\ln \frac{x}{zw} = \ln x - \ln(zw)$$

Quotient rule first.

$$= \ln x - \ln z - \ln w$$

Then the product rule, and the minus reaches both.

$$\ln \sqrt{x^5(x-2)^8 \div (x-3)^3} \rightarrow \frac{1}{3} [5 \ln x + 8 \ln(x-2) - \ln(x-3)]$$

A cube root is a power of one third, so the whole bracket is scaled by it.

## Simplifying

### WORKED EXAMPLES

$$\ln e^{3x} = 3x$$

The two undo each other.

$$\log 1 + \log 1000 = 0 + 3 = 3$$

$$1000 = 10^3.$$

$$\log_7 {}^9\sqrt{7^8} = \frac{8}{9}$$

$$\text{Because } {}^9\sqrt{7^8} = 7^{8/9}.$$

$$\log_3 \frac{27}{81} = \log_3 \frac{1}{3} = -1$$

$$\frac{3^3}{3^4} = 3^{-1}.$$

$$\ln e + \log \frac{1}{10} = 1 + (-1) = 0$$

Both special values.

### In summary

$$\exp(\ln u) = u \quad \ln(\exp x) = x \quad \exp(x) = u \Leftrightarrow x = \ln u$$

$$a^x = \exp(x \ln a) \text{ for } a > 0$$

### WATCH OUT

- $\log(a + b)$  does not break up. There is no rule for the logarithm of a sum.
- $\frac{\log a}{\log b}$  is not  $\log(a - b)$ , and  $(\log a)^2$  is not  $2 \log a$ .
- Isolate the exponential before taking logarithms, and reject any answer that makes an original logarithm take a non-positive input.

## Week 2 review quiz

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Six questions in the Cyberlearn format. The operations, composition and inverses from this week, with the distractors the usual mistakes produce.

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**Quantitative Methods I.** Notes I wrote while learning this. Free to use and to share.

Student notes, not official course material. Nobody teaching the subject has written or checked them. Use them alongside your own notes and whatever your course gives you, not instead of it. Where this page and your class disagree, your class is right.